

Title: **Geothermal Project on TU Delft Campus – DEL-GT-01 and DEL-GT-02 Wireline Logs**

DOI of this dataset: <https://www.doi.org/10.4121/2a7b2a63-dd7b-46bc-a275-97729b3ab348>

Authors: Auke Barnhoorn, Andrea Vondrak, Susanne Laumann, Beer van Esser, Hemmo A. Abels, Phil Vardon, and Liliana Vargas Meleza

Affiliation: TU Delft, Faculty of Civil Engineering and Geosciences (CEG), Department of Geoscience and Engineering (GSE)

Corresponding author: Auke Barnhoorn auke.barnhoorn@tudelft.nl

DATA AVAILABILITY

Embargo period: From 2023-12-31 to 2025-12-31

Licence: CC BY 4.0

Note that the exact same set of wireline logs will be made publicly available (in 2028) by NLOG (*Nederlandse Olie- en Gasportal*):

DEL-GT-01: <https://www.nlog.nl/nlog-mapviewer/brh/3888610845?lang=nl>

DEL-GT-02: <https://www.nlog.nl/nlog-mapviewer/brh/3895797327?lang=nl>

DATA SET DESCRIPTION

An extensive wireline logging campaign was undertaken during the installation of the geothermal doublet on TU Delft Campus, between June and November of 2023. This large suite of open-hole well logs acquired in the reservoir section of both wells, being DEL-GT-01 the producer and DEL-GT-02 the injector, is rarely undertaken in geothermal projects, which makes this data set of **exceptional scientific value**.

The description of the data collection methods is provided in the **Drilling Report of the campus geothermal doublet** (DOI to be assigned).

Below are listed all wireline logs and the depth range. Note that there were some logs acquired in cased borehole during sidetrack 1 (DEL-GT-02-S1).

Well	Logging Type	Log Type	Start Depth (m)	End Depth (m)
DEL-GT-01	Open Hole	BOREHOLE PROFILE LOG	2507	2934
		ACOUSTIC CEMENT BOND LOG	17.93	2610
		COMPENSATED NEUTRON LOG	2561	2934
		COMPENSATED Z-DENSILOG	2561	2934
		GEOEXPLORER	2565.8	2926
		ULTRASONIC EXPLORER	2565.8	2926
		MAGNETIC RESONANCE & EXPLORER LOG	2561	2934
		SPECTRAL LOG	2536.54	2934
		X-MULTIPOLE ARRAY ACOUSTIC LOG	2536.54	2934
		HIGH DEFINITION INDUCTION LOG	2536.54	2934
DEL-GT-02-S1	Case Hole	HIGH DEFINITION INDUCTION LOG	819.5	939.8
		X-MULTIPOLE ARRAY ACOUSTIC LOG	819.5	939.8
		GAMMA RAY	28.26	860
		BOREHOLE PROFILE LOG	819.5	939.8
		ACOUSTIC CEMENT BOND LOG	28.26	860
DEL-GT-02-S2	Open Hole	BOREHOLE PROFILE LOG	2179.87	2587.17
		ACOUSTIC CEMENT BOND LOG	41.6	2200

COMPENSATED NEUTRON LOG	2208.79	2588.43
COMPENSATED Z-DENSILOG	2208.79	2588.43
GEOEXPLORER	2199.89	2575
ULTRASONIC EXPLORER	2199.89	2575
MAGNETIC RESONANCE & EXPLORER LOG	2232.73	2582.19
SPECTRAL LOG	2232.73	2582.19
X-MULTIPOLE ARRAY ACOUSTIC LOG	2179.87	2587.17
HIGH DEFINITION INDUCTION LOG	2179.87	2587.17

DATA SET CONTENTS

The ZIP file DELGT01_DELGT02_wireline_logs.zop contains the following data files:

- Composite log image files in PDF format, at scales of 1:200 or 1:1000
- Log data files in LAS Format version 2.0
- Log data files in DLIS format (including metadata files in .meta and .dlis.scan.txt formats)

DLIS files can be viewed with specialist logging software or with open-source python packages such as *dlisio* (<https://pypi.org/project/dlisio>). In addition, LAS files can also be viewed in all specialists logging software but also *Notepad* (Windows), *VI* (Unix) or spreadsheets (e.g. Microsoft Excel) are capable of manipulating an ASCII text file.

LIST OF ABBREVIATIONS

GT01	DEL-GT-01
GT02-S1	DEL-GT-02-S1
GT02-S2	DEL-GT-02-S2
R1	Run 1
R2	Run 2
R3	Run 3
BHP	BOREHOLE PROFILE LOG
CBL	ACOUSTIC CEMENT BOND LOG
CN	COMPENSATED NEUTRON LOG
ZDL	COMPENSATED Z-DENSILOG
GXPL	GeoXplorer
UXPL	Ultra Sonic Explorer
MREX	MAGNETIC RESONANCE & EXPLORER LOG
SL	SPECTRAL LOG
XMAC	X-MULTIPOLE ARRAY ACOUSTIC LOG
HDIL	HIGH DEFINITION INDUCTION LOG

FILE STRUCTURE

```
+---GT01
| +---R1_BHP
| | TU_DELFT_GT01_8.5in_WL_run11_09Sep2023_BHP.dlis
| | TU_DELFT_GT01_8.5in_WL_run11_09Sep2023_BHP.dlis.scan.txt
| | TU_DELFT_GT01_8.5in_WL_run11_09Sep2023_BHP.meta
| | TU_DELFT_GT01_8.5in_WL_run11_09Sep2023_BHP.PDF
| | TU_DELFT_GT01_8.5in_WL_run11_09Sep2023_BHP_MAIN.las
| |
| +---R1_CBL
| | TU_DELFT_GT01_8.5in_WL_run11_09Sep2023_CBL.dlis
| | TU_DELFT_GT01_8.5in_WL_run11_09Sep2023_CBL.dlis.scan.txt
| | TU_DELFT_GT01_8.5in_WL_run11_09Sep2023_CBL.las
| | TU_DELFT_GT01_8.5in_WL_run11_09Sep2023_CBL.meta
```

```

| | TU_DELFT_GT01_8.5in_WL_run11_09Sep2023_CBL.PDF
| |
| +---R1_SL
| | TU_DELFT_GT01_8.5in_WL_run11_09Sep2023_SL.dlis
| | TU_DELFT_GT01_8.5in_WL_run11_09Sep2023_SL.dlis.scan.txt
| | TU_DELFT_GT01_8.5in_WL_run11_09Sep2023_SL.meta
| | TU_DELFT_GT01_8.5in_WL_run11_09Sep2023_SL.PDF
| | TU_DELFT_GT01_8.5in_WL_run11_09Sep2023_SL_MAIN.las
| |
| +---R1_XMAC-HDIL
| | TU_DELFT_GT01_8.5in_WL_run11_09Sep2023_XMAC_HDIL.dlis
| | TU_DELFT_GT01_8.5in_WL_run11_09Sep2023_XMAC_HDIL.dlis.scan.txt
| | TU_DELFT_GT01_8.5in_WL_run11_09Sep2023_XMAC_HDIL.meta
| | TU_DELFT_GT01_8.5in_WL_run11_09Sep2023_XMAC_HDIL.PDF
| | TU_DELFT_GT01_8.5in_WL_run11_09Sep2023_XMAC_HDIL_MAIN.las
| |
| +---R2_GXPL_UXPL
| | TU_DELFT_GT01_8.5in_WL_run11_10Sep2023_UXPL_GXPL.dlis
| | TU_DELFT_GT01_8.5in_WL_run11_10Sep2023_UXPL_GXPL.dlis.scan.txt
| | TU_DELFT_GT01_8.5in_WL_run11_10Sep2023_UXPL_GXPL.meta
| | TU_DELFT_GT01_8.5in_WL_run11_10Sep2023_UXPL_GXPL.PDF
| | TU_DELFT_GT01_8.5in_WL_run11_10Sep2023_UXPL_GXPL_MAIN.las
| |
| +---R3_CN_ZDL
| | TU_DELFT_GT01_8.5in_WL_run31_10Sep2023_ZDL_CN.dlis
| | TU_DELFT_GT01_8.5in_WL_run31_10Sep2023_ZDL_CN.dlis.scan.txt
| | TU_DELFT_GT01_8.5in_WL_run31_10Sep2023_ZDL_CN.meta
| | TU_DELFT_GT01_8.5in_WL_run31_10Sep2023_ZDL_CN.PDF
| | TU_DELFT_GT01_8.5in_WL_run31_10Sep2023_ZDL_CN_MAIN.las
| |
| \---R3_MREX
| | TU_DELFT_GT01_8.5in_WL_run31_10Sep2023_MREX.meta
| | TU_DELFT_GT01_8.5in_WL_run31_10Sep2023_MREX.PDF
| | TU_DELFT_GT01_8.5in_WL_run31_10Sep2023_mrex_cbw_depth.las
| | TU_DELFT_GT01_8.5in_WL_run31_10Sep2023_mrex_pplo_main_depth.las
| | TU_DELFT_GT01_8.5in_WL_run31_10Sep2023_MREX_ZDL_CN.dlis
| | TU_DELFT_GT01_8.5in_WL_run31_10Sep2023_MREX_ZDL_CN.dlis.scan.txt
| |
| +---GT02_S1
| | +---R1_BHP
| | | TU_DELFT_GT-02-S1_12.25in_WL_run11_08Nov2023_BHP.meta
| | | TU_DELFT_GT-02-S1_12.25in_WL_run11_08Nov2023_BHP.PDF
| | | TU_DELFT_GT-02-S1_12.25in_WL_run11_08Oct2023_HDIL_XMAC_GR_BHP.dlis
| | | TU_DELFT_GT-02-S1_12.25in_WL_run11_08Oct2023_HDIL_XMAC_GR_BHP.dlis.scan.txt
| | | TU_DELFT_GT021-S1_12.25in_WL_run11_08Nov2023_BHP_MAIN.las
| | |
| | +---R1_CBL
| | | TU_DELFT_GT-02-S1_12.25in_WL_run11_08Nov2023_CBL.meta
| | | TU_DELFT_GT-02-S1_12.25in_WL_run11_08Nov2023_CBL.PDF
| | | TU_DELFT_GT-02-S1_12.25in_WL_run11_08Oct2023_CBL.dlis
| | | TU_DELFT_GT-02-S1_12.25in_WL_run11_08Oct2023_CBL.dlis.scan.txt
| | | TU_DELFT_GT021-S1_12.25in_WL_run11_08Nov2023_CBL.las
| | |
| | \---R1_XMAC_HDIL
| | | TU_DELFT_GT-02-S1_12.25in_WL_run11_08Nov2023_HDIL_XMAC.meta
| | | TU_DELFT_GT-02-S1_12.25in_WL_run11_08Nov2023_HDIL_XMAC.PDF

```

```

|      TU_DELFT_GT-02-S1_12.25in_WL_run11_08Oct2023_HDIL_XMAC_GR_BHP.dlis
|      TU_DELFT_GT-02-S1_12.25in_WL_run11_08Oct2023_HDIL_XMAC_GR_BHP.dlis.scan.txt
|      TU_DELFT_GT021-S1_12.25in_WL_run11_08Nov2023_XMAC_HDIL_MAIN.las
|
|---GT02_S2
|   +---R1_BHP
|   |   TU_DELFT_GT02_S2_8.5in_WL_run11_28Nov2023_BHP_field.dlis
|   |   TU_DELFT_GT02_S2_8.5in_WL_run11_28Nov2023_BHP_field.dlis.scan.txt
|   |   TU_DELFT_GT02_S2_8.5in_WL_run11_28Nov2023_BHP_field.meta
|   |   TU_DELFT_GT02_S2_8.5in_WL_run11_28Nov2023_BHP_field.PDF
|   |   TU_DELFT_GT02_S2_8.5in_WL_run11_28Nov2023_BHP__main_field.las
|   |
|   +---R1_CBL
|   |   TU_DELFT_GT02_S2_8.5in_WL_run11_28Nov2023_CBL_field.dlis
|   |   TU_DELFT_GT02_S2_8.5in_WL_run11_28Nov2023_CBL_field.dlis.scan.txt
|   |   TU_DELFT_GT02_S2_8.5in_WL_run11_28Nov2023_CBL_field.meta
|   |   TU_DELFT_GT02_S2_8.5in_WL_run11_28Nov2023_CBL_field.PDF
|   |   TU_DELFT_GT02_S2_9.625in_WL_run11_28Nov2023_CBL_field.las
|   |
|   +---R1_XMAC_HDIL
|   |   TU_DELFT_GT02_S2_8.5in_WL_run11_28Nov2023_XMAC_HDIL_field.meta
|   |   TU_DELFT_GT02_S2_8.5in_WL_run11_28Nov2023_XMAC_HDIL_field.PDF
|   |   TU_DELFT_GT02_S2_8.5in_WL_run11_28Nov2023_XMAC_HDIL_WGI_field.dlis
|   |   TU_DELFT_GT02_S2_8.5in_WL_run11_28Nov2023_XMAC_HDIL_WGI_field.dlis.scan.inp
|   |   TU_DELFT_GT02_S2_8.5in_WL_run11_28Nov2023_XMAC_HDIL_WGI_Main.las
|   |
|   +---R2_GXPL_UXPL
|   |   TU_DELFT_GT02_S2_8.5in_WL_run21_28Nov2023_GXPL-UXPL_field.dlis
|   |   TU_DELFT_GT02_S2_8.5in_WL_run21_28Nov2023_GXPL-UXPL_field.dlisscan.txt
|   |   TU_DELFT_GT02_S2_8.5in_WL_run21_28Nov2023_GXPL-UXPL_field.meta
|   |   TU_DELFT_GT02_S2_8.5in_WL_run21_28Nov2023_GXPL-UXPL_field.PDF
|   |   TU_DELFT_GT02_S2_8.5in_WL_run31_29Nov2023_GXPL-UXPL_main_field.las
|   |
|   +---R3_CN_ZDL
|   |   TU_DELFT_GT02_S2_8.5in_WL_run31_29Nov2023_CN_ZDL_field.dlis
|   |   TU_DELFT_GT02_S2_8.5in_WL_run31_29Nov2023_CN_ZDL_field.dlis.scan.txt
|   |   TU_DELFT_GT02_S2_8.5in_WL_run31_29Nov2023_CN_ZDL_field.meta
|   |   TU_DELFT_GT02_S2_8.5in_WL_run31_29Nov2023_CN_ZDL_field.PDF
|   |   TU_DELFT_GT02_S2_8.5in_WL_run31_29Nov2023_DSL_CN_ZDL_Main_p850b_XC302.las
|   |
|   +---R3_MREX
|   |   TU_DELFT_GT02_S2_8.5in_WL_run31_29Nov2023_DSL_CN_ZDL_MREX_field.dlis
|   |
|   TU_DELFT_GT02_S2_8.5in_WL_run31_29Nov2023_DSL_CN_ZDL_MREX_Main_p850b_XC304.las
|   |   TU_DELFT_GT02_S2_8.5in_WL_run31_29Nov2023_MREX_field.meta
|   |   TU_DELFT_GT02_S2_8.5in_WL_run31_29Nov2023_MREX_field.PDF
|   |
|   |---R3_SL
|   |   TU_DELFT_GT02_S2_8.5in_WL_run31_29Nov2023_SL_field.dlis
|   |   TU_DELFT_GT02_S2_8.5in_WL_run31_29Nov2023_SL_field.dlis.scan.txt
|   |   TU_DELFT_GT02_S2_8.5in_WL_run31_29Nov2023_SL_field.meta
|   |   TU_DELFT_GT02_S2_8.5in_WL_run31_29Nov2023_SL_field.PDF
|   |   TU_DELFT_GT02_S2_8.5in_WL_run31_29Nov2023_SL_main_field.las

```